BITS, PILANI – HYDERABAD CAMPUS
BITS F464 MACHINE LEARNING
SECOND SEMESTER 2025-2026

Assignment 2
Automated Machine Learning Pipeline and Dashboard for Clinical Prediction under Temporal Shift in EHR Data

Dataset Description

The dataset used in this assignment is a structured healthcare dataset generated using a synthetic electronic health record (EHR) simulation framework. It is designed to support data-driven analysis and predictive modeling in clinical settings by capturing the relationships between patient demographics, clinical observations, and diagnosed medical conditions across diverse patient populations. Each record in the dataset corresponds to patient-level or encounter-level information collected over time and includes detailed attributes related to physiological measurements, demographic characteristics, and health status.

Patient-related attributes describe demographic and socio-economic characteristics such as age, gender, geographic location, and income. Clinical observation features quantify a wide range of physiological measurements, including vital signs and laboratory test results, reflecting the patient's health condition at different points in time. Condition-related data capture diagnosed diseases using standardized medical codes and descriptive labels, enabling the identification of clinically relevant outcomes.

The dataset provides detailed clinical indicators derived from routine healthcare interactions, including measurements such as body height, body weight, and other assessment scores, along with associated metadata, including units and observation categories. In addition to baseline patient information, the dataset includes longitudinal clinical observations across multiple encounters, enabling the aggregation and analysis of patient-specific trends and patterns.

Designed with scalability and interoperability in mind, the dataset follows a multi-table relational structure, where different aspects of patient information are stored in separate but linked tables. This structure enables integration of heterogeneous data sources into a unified analytical framework. The dataset supports advanced analytical workflows, including feature engineering, predictive modeling, and evaluation of machine learning models for healthcare applications, making it suitable for benchmarking clinical prediction tasks and studying data-driven decision-making in healthcare systems.

Link for dataset:
https://drive.google.com/drive/folders/1d7QielEDfhua8YfU77U0EmPF048LEcLv

By completing this assignment, students will be able to

- Integrate and preprocess multi-source real-world data, including handling relational datasets and aggregating features.
- Perform exploratory data analysis using statistical methods and visualizations to uncover patterns in healthcare data.
- Design and implement classification models, including Decision Trees, Support Vector Machines, and Neural Networks.
- Evaluate and compare model performance using appropriate classification metrics and interpret the results.
- Analyze the impact of model complexity on bias, variance, and generalization performance.
- Develop a reproducible machine learning pipeline and understand the role of automation and continual learning in model development workflows under temporal data shifts.
- ***"Students may choose different subsets of observations or aggregation strategies based on data availability."***

Tasks to be done

1. **Dashboard Development and Deployment:** Design and build an interactive dashboard using Streamlit (or a similar framework of your choice) to showcase the pipeline's functionality, model performance, and key data insights.
2. **Data Processing, Feature Engineering and Splitting:**
    a. Load the multi-table dataset and merge the CSV files based on the patient identifier.
    b. Feature Engineering: Transform raw clinical observations into meaningful features through aggregation (e.g., mean, variance). Apply encoding techniques for categorical variables. Normalize or standardize features where required.
    c. Filter the unified dataset into two distinct temporal datasets: **Dataset 1 (Historical)** and **Dataset 2 (Current)**, based on a specific timestamp or date column.
    d. Perform comprehensive preprocessing and Exploratory Data Analysis (EDA) on both Dataset 1 and Dataset 2.
    e. Create separate train/test splits for both Dataset 1 and Dataset 2.
3. **Model Training and Cross-Dataset Evaluation:**
    a. Train the classification models mentioned below (Decision Tree, SVM, Neural Network) using the training set of **Dataset 1**. Save the trained model configurations.
    b. Evaluate and show the performance of the Dataset 1 models on *both* the test set of Dataset 1 and the test set of Dataset 2.
    c. Include the evaluative metrics like Accuracy, Precision, Recall, F1-score, ROC curves.
    d. Model Complexity and Generalization: Compare training and testing performance to analyze underfitting and overfitting. Study how model complexity influences performance. Discuss bias-variance trade-offs across different models.

   e. Model Interpretation: Analyze feature importance for interpretable models such as Decision Trees. Discuss how different models utilize input features. Interpret results in the context of the prediction task.
   f. Analyze how different feature representations affect model performance.
4. **Continual Learning Implementation:**
   a. Implement a continual learning strategy on the train set of **Dataset 2**, leveraging the models trained on Dataset 1 as a starting point (e.g., using transfer learning or fine-tuning).
   b. Evaluate the continually learned model on the test set of **Dataset 2**.
5. **Reporting, Visualization, and Analysis:**
   a. Visualize all model performance comparisons, including the impact of the temporal shift (Task 3.b results) and the effectiveness of the continual learning approach (Task 4.b results).
   b. Analyze feature importance, model complexity, and discuss the implications of data drift and continual learning in this healthcare context.

Detailed Task Description

1. **Data Loading, Splitting, and Preprocessing**

   Load the datasets using Pandas and inspect their structure. Integrate the multiple tables using the patient identifier. Define a specific timestamp to split the data into **Dataset 1 (Historical)** and **Dataset 2 (Current)**. Check for missing values and perform comprehensive feature engineering, including transforming raw clinical observations into meaningful features through aggregation (e.g., mean, variance), applying encoding techniques for categorical variables, and normalizing or standardizing features where required. Handle missing values appropriately for both datasets. **Justify the selection of input features and define the classification target (medical condition).**

2. **Exploratory Data Analysis (EDA) and Train/Test Split**

   Perform descriptive statistics, visualizations, and class distribution analysis for *both* Dataset 1 and Dataset 2. Summarize key observations and compare the two datasets to identify potential data drift over time. Create appropriate train and test splits for both datasets.

3. **Model Training and Evaluation**

   Train the following classification models using Dataset 1's train set: Decision Tree, Support Vector Machine (SVM), and Neural Network (MLP). Apply appropriate preprocessing (e.g., feature scaling). Evaluate models using standard metrics (Accuracy, Precision, Recall, F1-score) and confusion matrices. *Crucially*, evaluate the performance of these models on *both* Dataset 1's test set and Dataset 2's test set to assess generalization across time.

   a. **Model Complexity and Generalization:** Compare training and testing performance to analyze underfitting and overfitting. Study how model complexity influences performance and discuss bias-variance trade-offs across different models.

b. **Model Interpretation:** Analyze feature importance for interpretable models (e.g., Decision Trees). Discuss how different models utilize input features. Analyze how different feature representations affect model performance and interpret results in the context of the prediction task.

4. **Continual Learning**

   Using the trained models from Dataset 1 as initial checkpoints, apply a suitable continual learning technique (e.g., fine-tuning the model with Dataset 2's training data). Evaluate the final continually learned model using Dataset 2's test set.

5. **Dashboard and Video Reporting**

   Develop a functional dashboard that allows for interactive viewing of the final models' comparative performance and insights derived from the data. The final video should summarize all findings, including the analysis of feature importance, model complexity, bias-variance trade-offs, the influence of temporal filtering and feature representation, and conclude on the success of the continual learning strategy.

Submission Requirements

- Submit **a single zip file** named **TeamXX_Assignment2.zip** containing TeamXX_Assignment2_dashboard.py, TeamXX_Assignment2_video.mp4
- The video must include
  - Steps used along with explanation
  - Labeled plots
  - Screenshots of the dashboard showing different functionalities
  - Summary of all findings, model performance, and data insights.
- Include the dashboard script.

Submission Format

- File name TeamXX_Assignment2_dashboard.py, video.mp4
- Include team details at the top of the video and in the dashboard.
- Submit a single zip file named TeamXX_Assignment2.zip containing the two files.

Timelines

- Final submission (Zip file on Google classroom) – Apr 22nd, 2026 11:59 PM
- Final Demos (in class) - Apr 24th, 2026 onwards

Evaluation Rubric (40 Marks)

1. Data Preprocessing and Exploratory Data Analysis (EDA) **(5 Marks)**
2. Feature Engineering and Representation (**5 Marks)**
3. Classification Models **(5 Marks)**
4. Hyperparameter Tuning and Model Optimization, Complexity, Generalization, and

Interpretation **(5 Marks)**
5. Overall ML Pipeline and Automation **(5 Marks)**
6. Visualization and Video Reporting **(5 Marks)**
7. Code Demo and Viva **(10 Marks - Individual Assessment)**

Academic Integrity

- This is a team-based assignment, and all submitted work must be the original effort of the team members. All team members must be present for the final viva else we will not be able to conduct the viva.
- Use of standard numerical and plotting libraries (e.g., NumPy, Pandas, Matplotlib) is permitted unless otherwise specified.
- Any external references (textbooks, papers, documentation) must be clearly cited.
- Plagiarism, code sharing between teams, or reuse of online implementations will result in disciplinary action as per the institute policy.

Contact Information

For any queries, please contact the course Teaching Assistants (TAs) via email communications through any other channels will not be considered. Please include all the listed TAs in your email for clarification.

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References

1. **https://docs.streamlit.io/get-started**
2. **https://streamlit.io/gallery**